Name: _____

WORKSHEET

Projections

Consider the following vectors:

$\mathbf{a} = (3, -5)$ $\mathbf{b} = (-2, 6)$ $\mathbf{c} = (-1, -4)$ $\mathbf{d} = (4, 2)$ $\mathbf{e} = (-5, -7)$ $\mathbf{f} = (-6, 1)$

$\mathbf{g} = 4\mathbf{i} + 5\mathbf{j}$ $\mathbf{h} = 2\mathbf{i} - 5\mathbf{j}$ $\mathbf{k}\,\mathbf{i} - 5\mathbf{j}$ $\mathbf{l} = -3\mathbf{i} + 2\mathbf{j}$ $\mathbf{m} = -5\mathbf{i} - \mathbf{j}$ $\mathbf{n} = -4\mathbf{i} + 6\mathbf{j}$

For the following questions, answer correct to two decimal places where necessary.

1 Find the following:

a $\hat{\mathbf{b}}$

b $\hat{\mathbf{d}}$

c $\hat{\mathbf{a}}$

d $\hat{\mathbf{g}}$

e $\hat{\mathbf{h}}$

f $\hat{\mathbf{n}}$

© Cengage Learning Australia Pty Ltd 2013

2 Find the following projections.

a **e** onto **a**

b **f** onto **b**

c **h** onto **m**

d **n** onto **g**

e **d** onto **c**

f **l** onto **k**

3 Sketch the following projections.

a The projection 13° of the vector (6, 52°)

b The projection 117° of the vector (9, 135°)

© Cengage Learning Australia Pty Ltd 2013

c The projection 224° of the vector (13, 300°)

d The projection 342° of the vector (8, 289°)

4 Evaluate the projections in question **3**.

a

b

c

d

5 Sketch the following projections.

a The projection 242° of the vector (12, 25°)

b The projection 36° of the vector (4, 152°)

© Cengage Learning Australia Pty Ltd 2013

c The projection 145° of the vector (6, 271°)

d The projection 288° of the vector (5, 105°)

6 Evaluate the projections in question **5**.

a

b

c

d

7 Find the projection of:

a 217° of the vector (15, 165°)

b 56° of the vector (7, 216°)

c 323° of the vector (3, 187°)

d 106° of the vector (12, 142°)

© Cengage Learning Australia Pty Ltd 2013

Answers

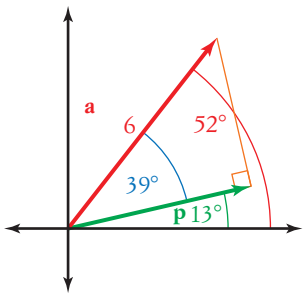
1 **a** $\left(\frac{-1}{\sqrt{10}}, \frac{3}{\sqrt{10}}\right)$ **c** $\left(\frac{3}{\sqrt{34}}, \frac{-5}{\sqrt{34}}\right)$ **e** $\left(\frac{2}{\sqrt{29}}, \frac{-5}{\sqrt{29}}\right)$

b $\left(\frac{2}{\sqrt{5}}, \frac{1}{\sqrt{5}}\right)$ **d** $\left(\frac{4}{\sqrt{41}}, \frac{5}{\sqrt{41}}\right)$ **f** $\left(\frac{-2}{\sqrt{13}}, \frac{3}{\sqrt{13}}\right)$

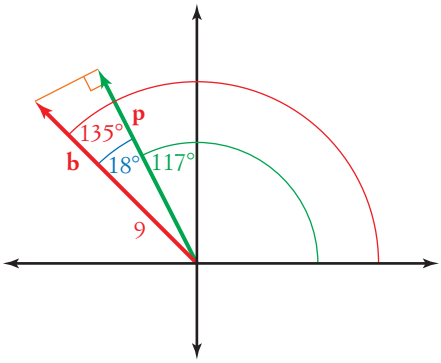
2 **a** $\frac{20}{\sqrt{34}} \approx 3.43$ **c** $-\frac{5}{\sqrt{26}} \approx -0.98$ **e** $-\frac{12}{\sqrt{17}} \approx -2.91$

b $\frac{18}{\sqrt{40}} \approx 2.85$ **d** $\frac{14}{\sqrt{41}} \approx 2.19$ **f** $-\frac{13}{\sqrt{26}} \approx -2.55$

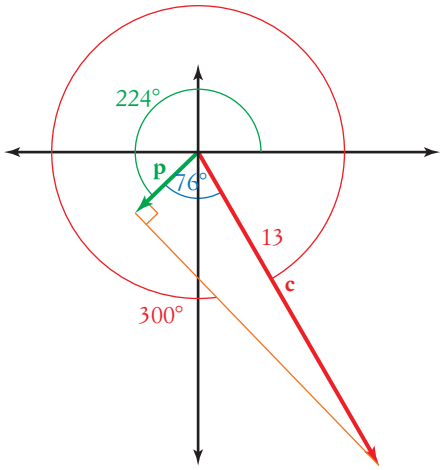
3 **a**



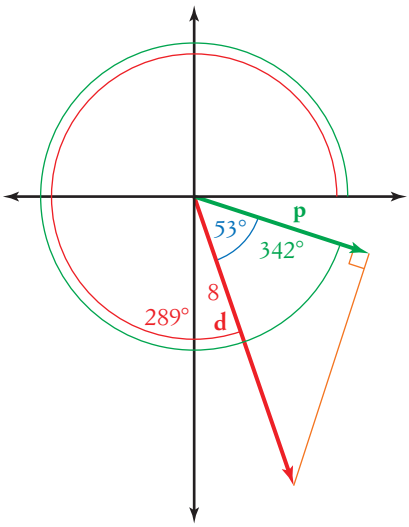
b



c



d



4 **a** 4.66

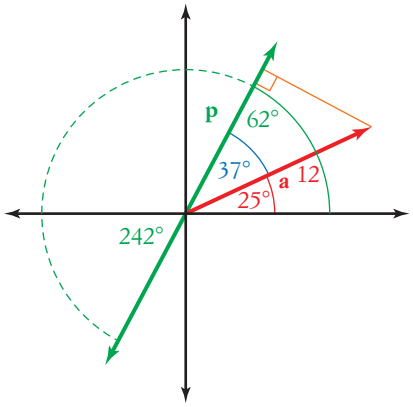
b 8.56

c 3.14

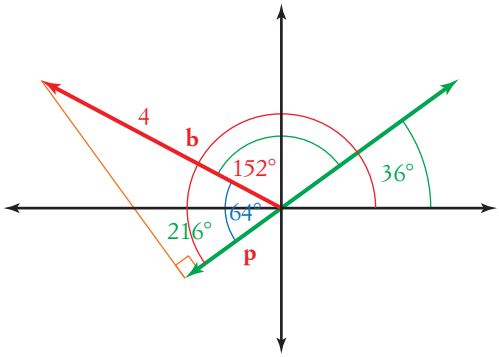
d 4.81

© Cengage Learning Australia Pty Ltd 2013

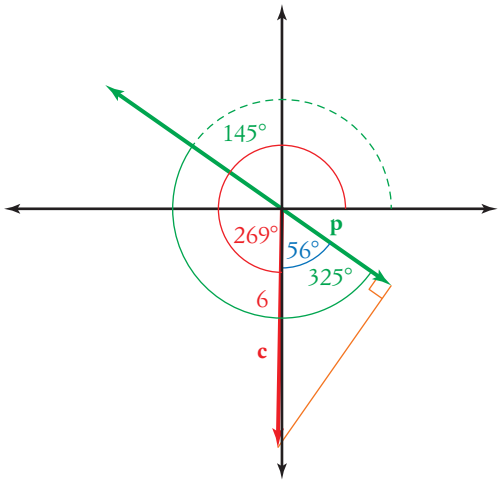
5 a



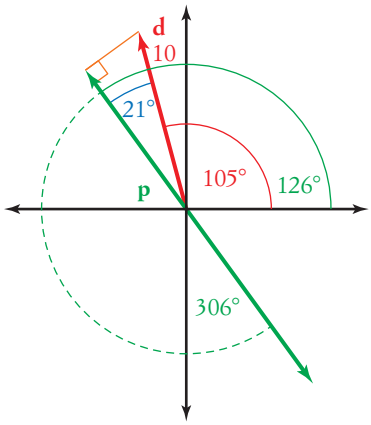
b



c



d



6 a −9.58

b −1.75

c −3.36

d −9.34

7 a 11.15

b −6.58

c −2.16

d 9.71

© Cengage Learning Australia Pty Ltd 2013